Harvard SEAS — Computer Science (PhD)

Statement of Purpose: Reliable Reinforcement Learning, Reasoning Geometry, and Fault-Tolerant Systems

Candidate:	Sham Satish Thakare (PRN: 72159719L, Seat: B191102032)
Target Program:	PhD in Computer Science, Harvard SEAS
Application Deadline:	December 15, 2026 (5:00 PM EST)
Current Affiliation:	Pre-Doctoral Researcher, Capital One Systems Research (Mentored by Yuri Shelke)
Academic Distinctions:	Gold Medal in Mathematics (2024), 8.70 CGPA (First Class with Distinction), 9.53 Final SGPA (Rank 1)
Target Faculty:	Prof. Sham Kakade (Kempner / SEAS), Prof. Boaz Barak, Prof. Minlan Yu, Prof. James Mickens

Statement of Purpose

Modern artificial intelligence systems increasingly operate as reasoning agents executing multi-step inference, autonomous tool orchestration, and distributed decision-making. However, the foundational mechanisms governing their reasoning stability, policy credit assignment, and fault tolerance remain brittle and poorly understood. Over the past two years as a Pre-Doctoral Researcher at Capital One Systems Research, mentored by Distinguished Engineer Yuri Shelke, I have investigated these foundational questions at the intersection of statistical machine learning theory and distributed systems architecture. At the Harvard John A. Paulson School of Engineering and Applied Sciences (SEAS), I aim to investigate the mathematical and algorithmic foundations of reliable reinforcement learning (RLVR), reasoning trajectory geometry, and provably robust distributed consensus for autonomous computing systems.

My research trajectory is grounded in rigorous mathematical training—recognized with the Gold Medal in Mathematics (2024) and an undergraduate CGPA of 8.70 / 10.00 (First Class with Distinction, Final Year SGPA 9.53 Rank 1) from Savitribai Phule Pune University. During my pre-doctoral research, I investigated the mathematical and empirical validity of using token predictive entropy as an uncertainty proxy in Group Relative Policy Optimization (GRPO). In reinforcement learning with verifiable rewards (RLVR), standard heuristics weight policy updates by token uncertainty. Through extensive controlled ablations on complex mathematical reasoning benchmarks (GSM8K, MATH), I discovered that token entropy is heavily confounded by sequence length, leading policy gradients to penalize verbose correct proofs rather than genuine epistemic hesitation. To resolve this, I formulated length-normalized trajectory metrics and counterfactual evaluation frameworks (recovery_eval, stateshift), demonstrating that true self-correction capability in reasoning models must be measured through state-contrast continuation: D_recovery = E[Δ_recovery] - E[Δ_control]. This formulation isolates genuine error recovery from unconditional generative capacity, providing a mathematically sound objective for policy optimization.

In parallel with my work on reasoning geometry, I have designed and implemented high-performance distributed systems in modern C++20. At Capital One Systems Research, I engineered quorumshift, an

adaptive Raft consensus engine that introduces dynamic quorum re-weighting under asymmetric network partitions. In cloud deployments spanning multiple regions, asymmetric degradation often causes standard Raft clusters to suffer catastrophic tail-latency spikes due to leader oscillations. By incorporating an analytical clock-drift bounded protocol (Δ-bounded drift), quorumshift maintains quorum intersection invariants while dynamically routing heartbeats and append-entries around degraded links, eliminating p99 latency spikes and ensuring deterministic state machine replication.

Harvard SEAS, in synergy with the Kempner Institute for the Study of Natural and Artificial Intelligence, provides an unparalleled ecosystem for my doctoral research. I am eager to work with Prof. Sham Kakade, whose foundational work on policy optimization, reinforcement learning theory, and statistical foundations of reasoning directly aligns with my vision for provably aligned and robust reasoning models. I am also deeply interested in collaborating with Prof. Boaz Barak on the theoretical capabilities and limits of deep learning and multi-step reasoning models. Furthermore, the pioneering systems research of Prof. Minlan Yu in high-throughput networked systems and Prof. James Mickens in distributed state machine fault tolerance offers an exceptional environment to integrate rigorous theoretical proofs with resilient, large-scale systems deployment.

Equipped with theoretical depth, industrial research discipline, and open-source systems provenance, I am eager to push the frontiers of robust artificial intelligence and distributed computing at Harvard SEAS.